Citi Bike Station Density and

Neighborhood Mobility in New York City

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CYPLAN 101 - Track B

Part 1: The Research Question

How did observed Citi Bike station density and usage vary with median household income and the share of households without a vehicle across New York City census tracts in September 2025, both citywide and within each borough?

NYC DOT announced plans to expand Citi Bike in the Bronx, Brooklyn, and Queens beginning in fall 2025, linking the expansion to equitable access [1]. This makes the distribution of stations a relevant planning question: do neighborhoods with fewer financial resources or less access to a private vehicle have comparable station coverage? We will investigate this question without assuming that a disparity exists.

We will compare station density with neighborhood income and vehicle availability across NYC, then repeat the comparisons within each borough. This helps show whether a citywide relationship also appears locally, while separating station coverage from usage where stations are present. A household without a vehicle is relevant to transportation planning, but vehicle availability alone does not establish a need or preference for bike share.

Scope: The study covers all five NYC boroughs: the Bronx, Brooklyn, Manhattan, Queens, and Staten Island. All eligible residential census tracts will enter the coverage analysis, including those with no observed stations. Usage comparisons will include only tracts with observed stations. New Jersey stations are outside the study area. Tracts support consistent demographic joins but are statistical areas, not exact neighborhood boundaries.

Target population: The population of interest is NYC residents and households, especially those in lower-income areas and those without a vehicle. Trip records describe activity at NYC stations, which may include visitors and commuters. They do not identify riders' residences, household incomes, or vehicle ownership.

Time frame: We will use trips starting September 1-30, 2025, with 2020-2024 ACS five-year estimates as demographic context. A single complete month keeps the analysis bounded and permits daily comparisons. It is a historical snapshot during the period of planned network expansion, not an evaluation of the expansion's effects or a measure of annual ridership.

Intended outcome: The project will quantify citywide and within-borough differences in observed station density and usage, and identify tracts that warrant closer assessment of coverage. We will produce a reproducible Python workflow, borough comparisons, and two interactive HTML maps with animated daily activity layers, presented on a GitHub Pages webpage for Track B. Recommendations will identify questions for further planning review rather than claim that the data determine optimal station locations.

Part 2: Data Description

We will combine three public data sources. Citi Bike provides the trip observations; ACS describes the surrounding households; TIGER/Line supplies the boundaries and land areas needed to compare tracts.

Citi Bike Trip Histories. We will use September 2025 trip records, filtered to NYC station locations, to identify observed stations, count departures, and examine daily activity and member/casual categories.

U.S. Census American Community Survey (ACS). ACS estimates describe median household income, households without a vehicle, total households, and population. Income will use B19013_001E (2024 inflation-adjusted dollars). The no-vehicle household share will be B08201_002E divided by B08201_001E. Population will use B01003_001E. We will retain the corresponding margins of error and annotations. These ACS figures represent 2020-2024, not conditions measured in September 2025.

U.S. Census TIGER/Line Census Tracts. We will use the New York tract file, filtered to the five NYC counties, to assign stations to tracts and boroughs, join ACS attributes by GEOID, and calculate land-area and population-normalized station density.

Dataset	Coverage	Granularity	Type	Format	Source
Citi Bike Trip Histories	NYC stations; September 2025	Individual trip	Tabular	CSV / ZIP	Citi Bike System Data [2]
American Community Survey	All NYC boroughs; 2020-2024	Census tract	Tabular	Text / API	U.S. Census Bureau ACS [3]
TIGER/Line Census Tracts	New York; 2024	Tract polygon	Vector	Shapefile	U.S. Census Bureau TIGER/Line [4]

Data acquisition. The monthly Citi Bike archive is approximately 1.03 GB compressed and may contain several CSVs; we will read every CSV in chunks. ACS will be downloaded from the official table-based Summary File, or queried with a Census API key. Source files, download dates, table definitions, and cleaning decisions will be recorded.

Terms and attribution. Citi Bike data is subject to the NYCBS Data Use Policy [5]. We will publish code and aggregated findings, provide official download links, and avoid republishing the raw archive as a standalone dataset. Census-produced data is generally not copyright protected in the United States [6]. All sources will be attributed.
Sources accessed October 4-5, 2026.

Part 3: Methods & Techniques

All acquisition, cleaning, analysis, and mapping will use Python, primarily pandas, GeoPandas, and Folium. The analysis will use spatial joins, aggregation, rates, group comparisons, and descriptive rank correlations.

GIS-based Spatial Analysis

- We will parse timestamps, retain September starts, preserve station IDs as strings, and remove duplicate ride IDs. We will audit missing IDs, coordinates, and times by date and user category. A missing destination will not remove an otherwise valid departure. Per-station usage will require a valid start station ID; coordinate-only starts will be reported separately. Stations will be identified from both endpoints using unique IDs and modal coordinate pairs; inconsistent locations will be reviewed.
- We will align station points and tracts in a projected coordinate system appropriate for NYC, spatially join stations to tracts and boroughs, and merge ACS data by the 11-digit GEOID. Residential comparisons require positive household counts, population, and land area. Tracts missing a grouping variable will remain mapped but be excluded from that comparison. Each station will be assigned to at most one tract. Departures will be assigned to the start station's NYC tract, counting interborough trips once. NYC destinations can also identify observed stations.
- For each tract, observed station density will equal distinct stations recorded during the month divided by land area in square kilometers. We will also calculate stations per 10,000 residents. Usage will equal September departures divided by observed stations. Tracts with no observed stations receive zero density and undefined usage; a borough with no observed stations remains in coverage comparisons. These measures do not represent dock capacity, real-time bike availability, or trips made by residents.

Comparative Neighborhood Analysis

- We will compare median density and usage across citywide quartiles of tract income and, separately, the no-vehicle household share. We will repeat these comparisons within each borough using the same thresholds. Each group will show its tract count, share with no observed stations, and density distribution; usage includes only tracts with observed stations. Empty groups will be labeled. Scatterplots and descriptive Spearman correlations will show citywide and within-borough relationships; undefined correlations will be marked not applicable. We will report thresholds and compare land-area and population normalization.

We will reconcile tract departure totals with retained NYC records, check totals by borough, verify that joins do not duplicate stations, and check ACS denominators and missing-value codes. To assess partial-month operation, we will repeat usage comparisons using stations observed in both the first and last week; this is a sensitivity check, not proof of continuous operation. We will inspect nearby stations across tract boundaries and avoid interpreting low within-tract density as a direct measure of walking access.

One interactive HTML map will compare tract station density with income and vehicle availability, including an animated daily departure layer. A second will animate daily station departures. Both will offer borough filters and fixed scales across dates and boroughs. Boxplots and scatterplots will compare neighborhoods citywide and within boroughs. Zero and missing values will have distinct symbols; positive-density classes will use documented citywide quantiles. Figures will include units, sources, dates, captions, and colorblind-safe colors. We will publish a download script, pinned dependencies, and a notebook that runs from start to finish.

Part 4: Timeline & Milestones

Proposed responsibilities: Tim will handle data preparation and mapping; Ethan will handle spatial joins and analysis. Both will write and review the report.

1. Oct 8: Confirm the question, scope, and responsibilities. Owners: Tim and Ethan.
2. Oct 15: Download sources and complete the missing-data audit. Owner: Tim.
3. Oct 22: Join tracts, stations, and ACS; validate measures. Owner: Ethan.
4. Nov 2, noon: Submit the midterm with an early map and 3-5 charts. Owners: both.
5. Nov 16: Finish citywide and borough comparisons and sensitivity checks. Owner: Ethan.
6. Dec 6: Complete the interactive maps and webpage draft. Owner: Tim.
7. Dec 15, 11:59 pm: Finish interpretation, run all code, and submit. Owners: both.

Part 5: Potential Challenges & Mitigation Plan

1. **Data volume and missingness.** The large trip archive can exceed notebook memory. We will process selected columns in chunks, aggregate before mapping, and keep lightweight derived tables. We will document dropped or unlocatable records and compare their frequency across dates and rider categories. Valid coordinates can preserve some spatial analysis when IDs are missing, but such records will not create inferred stations.
2. **Incomplete observation of service.** A station absent from trip records may have been unused, closed, or missing from the published data. We will label the measure as observed station density and state that zero means no station was observed. Current real-time feeds will not be substituted for a historical inventory. Without operational records, low usage cannot establish low demand, unmet demand, or service reliability.
3. **Timing and uncertainty.** ACS estimates cover five years and carry sampling uncertainty. We will retain margins of error, flag imprecise or unavailable estimates, and avoid precise rankings when uncertainty is large. We will report results only for September 2025. Any station openings during the month, weather, holidays, and seasonal patterns may affect activity.
4. **Geography and ecological inference.** Tract boundaries can hide differences within neighborhoods and separate residents from nearby stations. We will display adjacent stations and acknowledge the modifiable areal unit problem. Income and no-vehicle

shares describe tracts, not individual riders. We will not infer that riders at a station share the income or vehicle access of nearby households.

5. **Interpretation and privacy.** Jobs, transit connections, cycling infrastructure, and land use can affect station placement and trips. Within-borough comparisons reduce reliance on citywide averages but do not control for these factors. Findings describe associations, not causes or the effects of adding stations. Low-coverage tracts will be candidates for further review, not a definitive investment ranking. We will publish aggregate results and make no attempt to identify riders.